BIOCHEMISTRY CUMULATIVE EXAM

February 27, 1999

Sonication was used to prepare phospholipid vesicles containing 1 mM NaCl and 200 mM KCl on the inside of the vesicles. The vesicles were then washed and suspended in a solution containing 200 mM NaCl and 1 mM KCl which was on the outside of the vesicles.

1. (35) Describe what will happen to the membrane voltage, $V_m = V_{in} - V_{out}$ as a function of time after valinomycin is added to the solution. Valinomycin is a carrier-type transporter with a high permeability for K^+, but a very low permeability for Na^+. Calculate the membrane voltage V_m after it has stabilized.

2. (35) Describe what will happen to the membrane voltage V_m as a function of time after sodamycin is added to the solution. Sodamycin is a channel-type transporter with a very high permeability for Na^+, but a very low permeability for K^+. Calculate the membrane voltage V_m after it has stabilized.

3. (30) Describe what will happen to the membrane voltage V_m as a function of time after equimycin is added to the solution. Equimycin is a channel-type transporter with very high and equal permeabilities for Na^+ and K^+. Calculate the membrane voltage V_m after it has stabilized.

Possibly useful equations:

$\Delta G(X_{out} \rightarrow X_{in}) = RT\ln([X_{in}]/[X_{out}]) + ZF(V_{in} - V_{out})$. (Z is charge on X)
$\Delta G = \Delta G^o + RT\ln([\text{products}]/[\text{reactants}])$. $\Delta G^o = -nF\Delta E_o$
$R = 2$ cal/degree.mol, $F = 23$ kcal/volt.mol